

FORMULA MANIPAL DRIVERLESS

EC2 WORKSTATION DOCUMENTATION

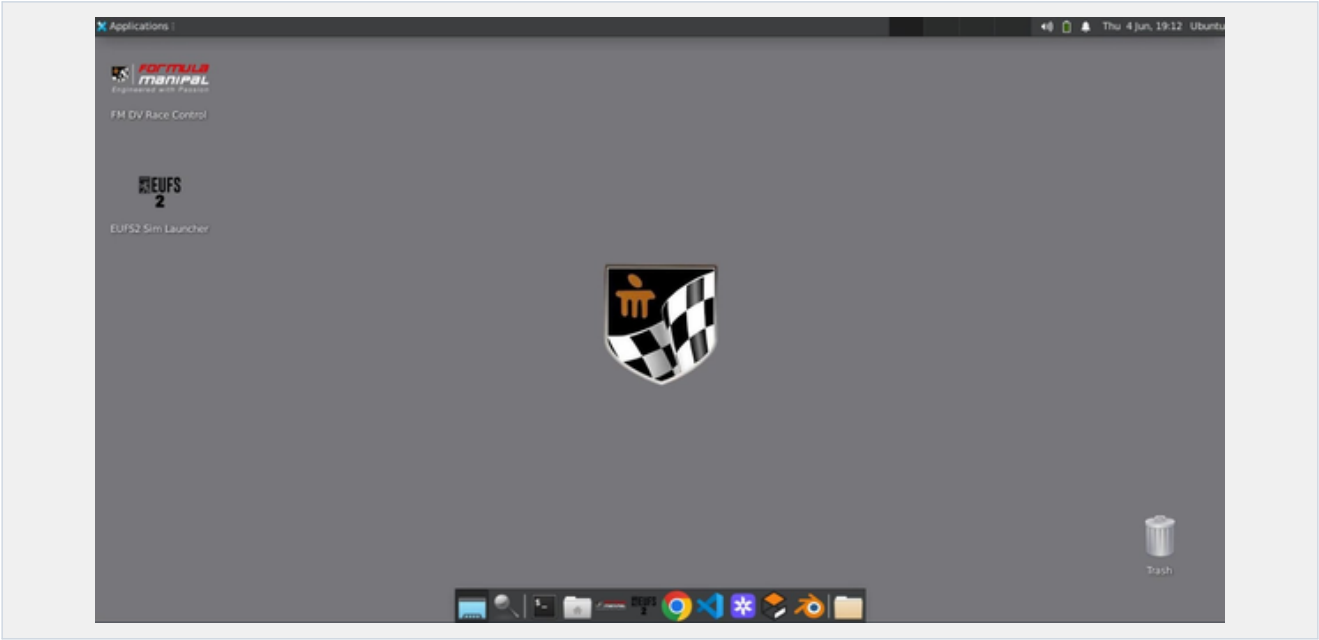


Figure 1 — The DCV remote desktop showing FM DV Race Control and EUFS2 Sim Launcher shortcuts

This document covers the complete setup, configuration, and day-to-day usage of the Formula Manipal Driverless AWS EC2 GPU workstation. It includes instructions for starting DCV sessions, workspace navigation, CARLA Simulator, EUFS Sim 2, Foxglove, and the full initial system setup from a fresh Ubuntu instance.

Instance Quick Reference

EC2 Public IP	13.204.88.75
DCV Web UI	https://13.204.88.75:8443
SSH User	ubuntu
OS	Ubuntu 22.04 LTS
GPU	NVIDIA (driver 550)
CARLA Version	0.9.16

ROS 2 Version	Humble
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A. STARTING A DCV SESSION

NICE DCV provides a GPU-accelerated remote desktop to the EC2 instance. Follow these steps after every reboot or fresh login.

Step 1 — Fix PEM File Permissions (Windows Only)

Run the following commands in an **Administrator Command Prompt**:

```
# Remove inherited permissions and grant read-only to current user
icacls C:\path\to\workstation.pem /inheritance:r
icacls C:\path\to\workstation.pem /grant:r %USERNAME%:R

# (Optional) Verify
icacls C:\path\to\workstation.pem
```

■ Linux/macOS users: run `chmod 400 workstation.pem`

Step 2 — SSH Into the Instance

```
ssh -i C:\path\to\workstation.pem ubuntu@

# Current instance:
ssh -i C:\Keys\workstation.pem ubuntu@13.204.88.75
```

Step 3 — Start / Verify DCV Server

```
sudo systemctl start dcvserver
sudo systemctl enable dcvserver
sudo systemctl status dcvserver

# List sessions
dcv list-sessions

# Create session if none exists
sudo dcv create-session --owner=ubuntu --type=virtual my-session
```

DCV Username	ubuntu
DCV Password	fmdv2026

Step 4 — Connect to the Remote Desktop

```
# Browser or DCV Client
https://13.204.88.75:8443/#my-session
```

■ Accept the self-signed TLS certificate warning — expected for an internal workstation.

B. OUR WORKSPACE

The remote desktop is an Xfce environment on Ubuntu. The taskbar at the bottom provides quick access to the most commonly used tools.

Desktop Overview

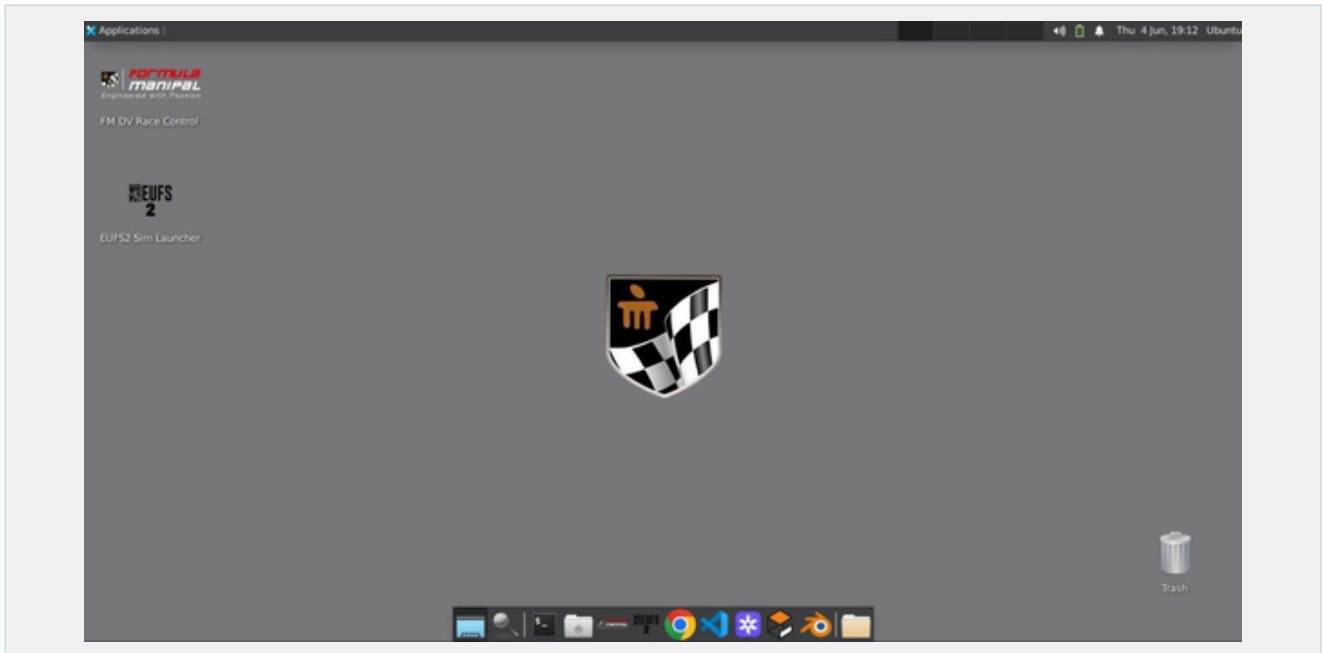


Figure 2 — The remote desktop. FM DV Race Control (top-left) and EUFS2 Sim Launcher shortcuts are visible. The taskbar at the bottom contains quick-launch icons.

Desktop Taskbar — Quick Access Items

- **Show Desktop** — Minimise all windows and reveal the desktop.
- **Search / Find Menu** — Launch any installed application by name.
- **Ubuntu Terminal** — Opens a Bash terminal in the home directory.
- **Folders** — File manager, opens to the ~/ home directory.
- **FM DV Race Control** — Launches CARLA + the Race Control panel.
- **EUFS2 Sim Launcher** — Launches EUFS2, Foxglove Bridge, and Foxglove Studio.
- **Google Chrome** — Pre-logged-in browser for GitHub / Drive.
- **VS Code** — Code editor.
- **Foxglove Studio** — ROS 2 data visualisation.
- **Blender** — 3D modelling for car FBX/STL assets.
- **Files** — Quick file manager access.

Workspace Folder Structure (~/Workspace/)

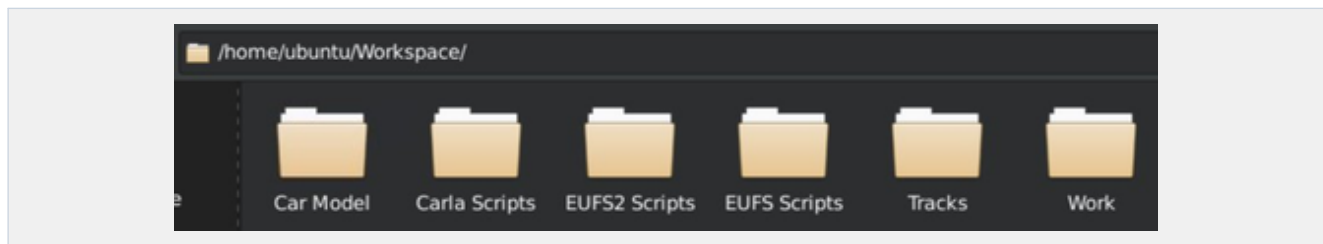


Figure 3 — /home/ubuntu/Workspace/ folder layout

Folder	Contents / Purpose
Car Model	FBX and STL files for the EUFS car and Formula Manipal car.
Carla Scripts	Python scripts for CARLA. Includes a sample compliant controller script.
EUFS2 Scripts	Python scripts for EUFS Sim 2. Includes a sample MPC script with all topics.
EUFS Scripts	Python scripts for the original EUFS simulator.
Tracks	csv_to_waypoints.py plot_waypoints.py transform_cones.py (WIP)
EUFS Tracks	All known track cone coordinates and corresponding waypoints.
Work	Sub-folders: Perception, Path Planning, SLAM, Simulation. Develop here first.

■ ■ The ~/Launchers/ folder must not be modified — it contains the one-click launcher scripts.

FM DV Race Control (CARLA Interface)

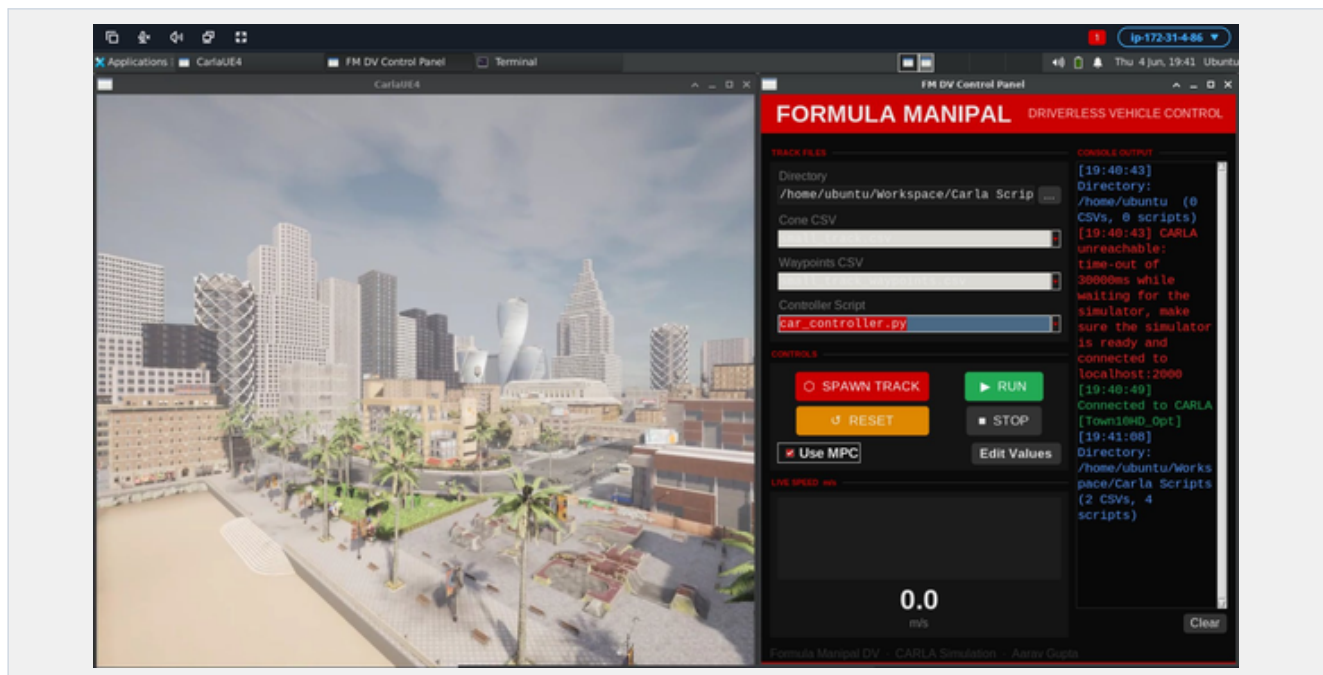


Figure 4 — FM DV Race Control panel alongside CARLA running the Town10HD_Opt map

The **FM DV Race Control** shortcut launches CARLA together with a custom control panel:

- A directory browser to select scripts, waypoints, and cone-coordinate CSVs.
- Buttons to **Spawn Track**, **Run**, **Reset**, and **Stop** the simulation.
- An **Edit Values** panel to tune algorithm parameters at runtime.
- A live console showing timestamped output from CARLA and the controller.

■■ Controller scripts must follow the specific format expected by Race Control. A compliant sample is in *Carla Scripts/*.

EUFS2 Sim Launcher

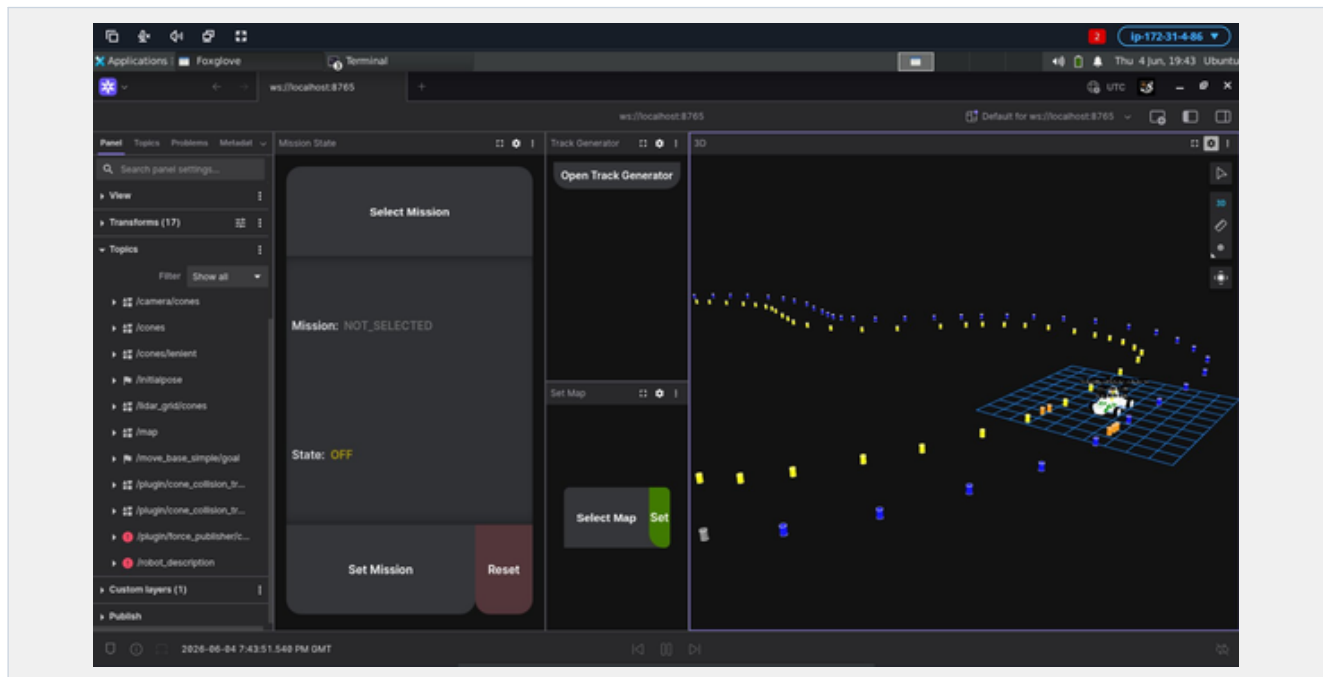


Figure 5 — Foxglove Studio connected to EUFS2 via WebSocket (ws://localhost:8765). Left panel shows ROS topics; right panel shows 3D cone/track visualisation.

The **EUFS2 Sim Launcher** opens three terminals simultaneously:

Terminal	Process	Notes
1	EUFS Sim 2	Full ROS 2 simulation environment.
2	Foxglove Bridge	Exposes ROS topics over WebSocket on port 8765.
3	Foxglove Studio	GPU-accelerated visualisation UI (auto-connects).

Once Foxglove opens, click **Open Connection** → **Foxglove WebSocket** → **Open** and confirm the URL is ws://localhost:8765. Then run your car controller script in a fourth terminal.

C. AVAILABLE APPLICATIONS

Google Chrome

Logged in as **fmdriverless26@gmail.com**. Used to access GitHub and Google Drive. Use the PAT token below as the password when Git prompts for credentials over HTTPS.

VS Code

Launch from any terminal: `code .` opens VS Code in the current directory.

Foxglove Studio

Primary ROS 2 data visualisation tool. Connect to `ws://localhost:8765`.

Gazebo

Alternative robotics simulation and visualisation environment.

Blender

Create and edit skeletal rigs and FBX car models for import into Unreal Engine 4 (CARLA).

GitHub Username	fmdriverless26
GitHub Token (PAT)	ghp_3VU0g0jwvDYSEpbSutCBQ3swi90mbJ2M7tPr
Gmail Account	fmdriverless26@gmail.com

■ ■ Use the PAT as your password when Git asks for credentials over HTTPS.

D. INITIAL SYSTEM SETUP

These steps are only needed when provisioning a **fresh Ubuntu EC2 instance**. Skip if the instance is already configured.

Step 1 — Install Essential Packages

```
sudo apt install -y \  
git curl wget build-essential cmake \  
python3-pip python3-venv software-properties-common \  
htop unzip net-tools
```

Step 2 — Install NVIDIA GPU Driver

```
sudo apt install -y nvidia-driver-550  
sudo reboot  
  
# After reboot – verify  
nvidia-smi  
  
# Set NVIDIA as default OpenGL vendor  
echo 'export __GLX_VENDOR_LIBRARY_NAME=nvidia' >> ~/.bashrc  
source ~/.bashrc
```

Step 3 — One-Shot Application Install Script

Save as `install_apps.sh` and run with `bash install_apps.sh`.

```
#!/bin/bash  
  
# Google Chrome  
wget -q -O /tmp/chrome.deb  
https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb  
sudo apt install -y /tmp/chrome.deb && rm /tmp/chrome.deb  
  
# VS Code  
sudo snap install code --classic  
  
# Blender  
sudo snap install blender --classic  
  
# LibreOffice  
sudo apt install -y libreoffice  
  
# Gazebo 11  
sudo apt install -y gazebo11 libgazebo11-dev  
  
# Build tools & utilities  
sudo apt install -y htop git curl wget build-essential cmake python3-pip  
  
# Extra tools  
sudo apt install -y ncdu filelight bleachbit rclone p7zip-full gimp flameshot  
simplescreenrecorder  
  
# NVIDIA driver
```

```
sudo apt install -y nvidia-driver-550  
echo 'export __GLX_VENDOR_LIBRARY_NAME=nvidia' >> ~/.bashrc && source ~/.bashrc  
echo "All applications installed."
```

Step 4 — Verify Installations

```
google-chrome --version && code --version && blender --version  
libreoffice --version && gazebo --version && nvidia-smi
```

E. EUFS SIM 2 + FOXGLOVE INSTALLATION

EUFS Sim 2 is a ROS 2-based Formula Student driverless simulation environment. Foxglove Bridge exposes ROS topics over WebSocket; Foxglove Studio provides the visualisation front-end.

1. Install ROS 2 Humble

```
sudo apt install -y software-properties-common
sudo add-apt-repository universe -y && sudo apt update
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
-o /usr/share/keyrings/ros-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) \
signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] \
http://packages.ros.org/ros2/ubuntu $(lsb_release -cs) main" \
| sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
sudo apt update
sudo apt install -y ros-humble-desktop python3-colcon-common-extensions python3-rosdep
```

2. Set Up the EUFS Sim 2 Workspace

```
mkdir -p ~/modified_eufs_sim_v2 && cd ~/modified_eufs_sim_v2
git clone https://github.com/formula-manipal-driverless/modified_eufs_sim_v2.git
cd modified_eufs_sim_v2
```

3. Install Dependencies with rosdep

```
cd ~/modified_eufs_sim_v2
sudo rosdep init # ignore 'already exists' warning
rosdep update
rosdep install --from-paths . --ignore-src -r -y
```

4. Build the Workspace

```
cd ~/modified_eufs_sim_v2/eufs-master
colcon build --symlink-install
```

5. Source the Workspace

```
echo 'source /opt/ros/humble/setup.bash' >> ~/.bashrc
echo 'source ~/modified_eufs_sim_v2/eufs-master/install/setup.bash' >> ~/.bashrc
source ~/.bashrc
```

6. Install Foxglove Bridge

```
sudo apt update
sudo apt install -y ros-humble-foxglove-bridge
```

7. Install Foxglove Studio

```
sudo snap install foxglove-studio
```

8. (Optional) Install EUFS CLI

```
cd ~/modified_eufs_sim_v2/eufs_sim/eufs_cli
pip install --user -e ./eufsccli && pip install --user -e ./eufsininstall && pip install --user -e ./eufssim
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc && source ~/.bashrc
eufs update
```

9. Launch EUFS Sim 2 (Terminal 1)

```
source ~/modified_eufs_sim_v2/eufs-master/install/setup.bash
ros2 launch eufs_sim2 eufs_sim2.launch.py
# - or with CLI -
eufs sim run
```

10. Launch Foxglove Bridge (Terminal 2)

```
source ~/modified_eufs_sim_v2/eufs-master/install/setup.bash
ros2 launch foxglove_bridge foxglove_bridge_launch.xml use_foxglove_whitelist:=False
```

11. Launch Foxglove Studio (Terminal 3)

```
__GLX_VENDOR_LIBRARY_NAME=nvidia foxglove-studio \
--enable-gpu-rasterization --ignore-gpu-blocklist --use-angle=gl
```

One-Click Launcher Script

Save as `~/Launchers/EUFS2/run_eufs2.py` and make executable with `chmod +x ~/Launchers/EUFS2/run_eufs2.py`.

```
#!/usr/bin/env python3
import subprocess, time, os, signal, sys
workspace_path = "/home/ubuntu/modified_eufs_sim_v2/eufs_sim/eufs-master"
cli_home = "/home/ubuntu/modified_eufs_sim_v2/eufs_sim/eufs_cli"
processes = []

def signal_handler(sig, frame):
    for p in processes:
        try: p.terminate()
        except: pass
    sys.exit(0)

def launch_terminal(cmd):
    full_cmd = (
        f"export EUFS_MASTER={workspace_path} && "
        f"export EUFS_CLI_HOME={cli_home} && "
```

```
"export PATH=\"$HOME/.local/bin:$PATH\" && "  
"source /opt/ros/humble/setup.bash && "  
f"source {workspace_path}/install/setup.bash && "  
f"{cmd}; exec bash"  
)  
processes.append(subprocess.Popen(['gnome-terminal', '--', 'bash', '-c', full_cmd],  
preexec_fn=os.setsid))  
  
if __name__ == "__main__":  
    signal.signal(signal.SIGINT, signal_handler)  
    launch_terminal("eufs sim run"); time.sleep(5)  
    launch_terminal("ros2 launch foxglove_bridge foxglove_bridge_launch.xml  
use_foxglove_whitelist:=False"); time.sleep(3)  
    launch_terminal("__GLX_VENDOR_LIBRARY_NAME=nvidia foxglove-studio --enable-gpu-rasterization  
--ignore-gpu-blocklist --use-angle=gl")  
    print("All launched. Ctrl+C to quit.")  
    try:  
        while True: time.sleep(1)  
    except KeyboardInterrupt: pass
```

F. CARLA SIMULATOR INSTALLATION

CARLA is an open-source autonomous driving simulator built on Unreal Engine 4. The team uses version **0.9.16**.

Step 1 — Download and Extract CARLA

```
# Download CARLA_0.9.16.tar.gz from:  
# https://github.com/carla-simulator/carla/releases  
  
tar -xzf CARLA_0.9.16.tar.gz -C ~/   
mv ~/CARLA_0.9.16 ~/Carla
```

Step 2 — Make the Launcher Executable

```
cd ~/Carla  
chmod +x CarlaUE4.sh
```

Step 3 — Run CARLA

```
# Default launch  
./CarlaUE4.sh  
  
# High quality, windowed 1600x900  
./CarlaUE4.sh -quality-level=High -ResX=1600 -ResY=900 -windowed  
  
# Epic quality, windowed 1920x1080  
./CarlaUE4.sh -quality-level=Epic -ResX=1920 -ResY=1080 -windowed
```

■■ Full CARLA documentation: <https://carla.readthedocs.io/en/latest/>

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